

Albert Andersen

ACADEMY PORTFOLIO
Technical Case Studies

🔧 Physicist | HPC System Administrator | Fullstack Developer

✉ alberttandersen@gmail.com 📍 Jakarta / Depok, Indonesia 🔗 [linkedin.com/in/alberttandersen](https://www.linkedin.com/in/alberttandersen)

🐙 github.com/alberttandersen 🌐 alberttandersen.github.io

🔗 INTERDISCIPLINARY FOUNDATION & MOTIVATION

Quantum Physics

HPC Supercomputing

Web Engineering

"I bridge the gap between theoretical physics and applied technology. My journey spans researching quantum materials to architecting the High-Performance Computing clusters that make such research possible, and developing robust enterprise software. I am driven by the desire to build resilient infrastructures and innovative solutions that accelerate scientific and business advancements."

PROJECT // 01

Enterprise B2B E-Catalog & Smart RFQ System

📄 Context: Work assignment for PT Wisu Varia Analitika

👤 Role: Individual (Fullstack Developer & IT Support)

🌐 wisuvar.com

Summary: Rebuilt a B2B e-catalog portal using Laravel 11 and Filament PHP, featuring an automated Smart Request for Quotation (RFQ) system and multi-language support.

📈 MEASURABLE IMPACT

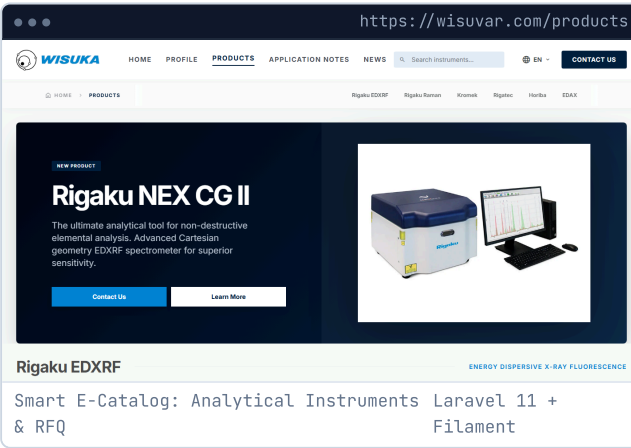
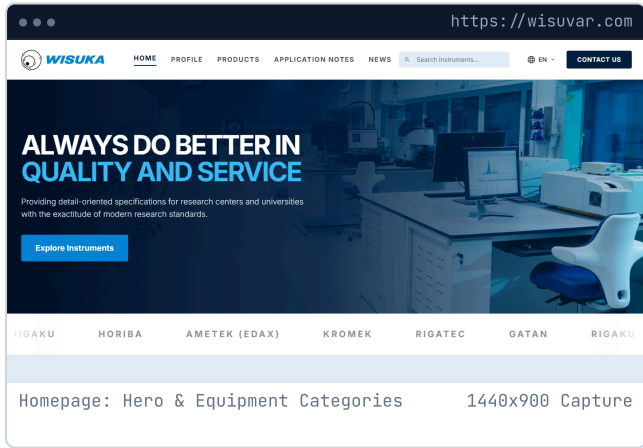
Accelerated client response times, blocked spam via SMTP Auto-Responder and IP Rate Limiting, and successfully recovered business email deliverability from spam boxes via strict DNS authentication (SPF, DKIM, DMARC).

💡 CORE LEARNINGS

Mastered the integration of real-time VPS performance monitoring into web dashboards and learned how to align technical DevOps solutions directly with business automation and SEO goals.

📷 AUTHENTIC PRODUCTION INTERFACE

✅ LIVE PRODUCTION DEPLOYMENT



PROJECT // 02

Centralized Research Web Services: JupyterHub & SSO

Context: Work assignment at PHOELabs & TCQM Lab Role: Individual (System Administrator)

jhub.tcqmui.my.id sso.tcqmui.my.id

Summary: Implemented and maintained remote access web services, including a centralized JupyterHub and a custom Single Sign-On (SSO) authentication flow for distributed user environments.

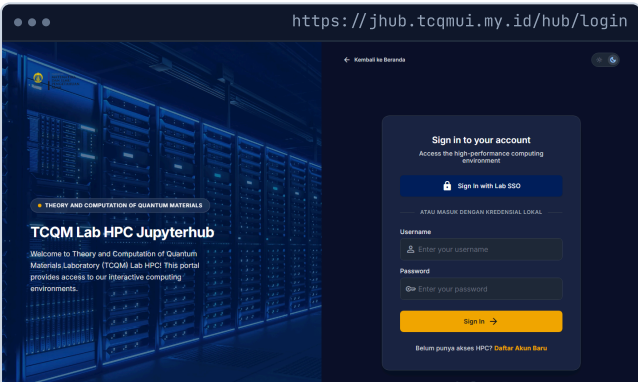
MEASURABLE IMPACT

Enabled secure, seamless remote access to lab computing resources, complete with NFS-based shared directories and custom access control, greatly improving operational flexibility for researchers.

CORE LEARNINGS

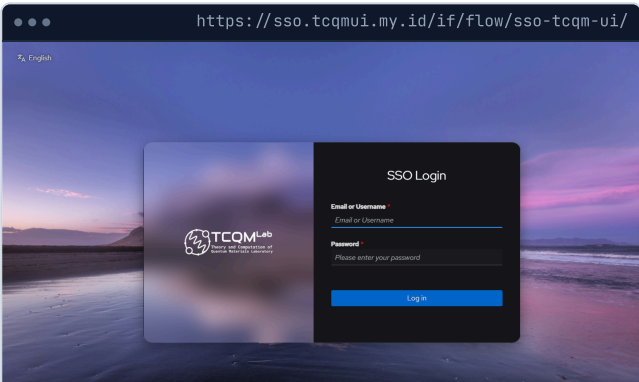
Gained hands-on expertise in deploying Nginx, configuring OAuth/OIDC protocols for SSO, and designing internal networks (multi-NIC, DHCP, multi-gateway routing) to connect master and compute nodes securely.

AUTHENTIC PRODUCTION INTERFACE



TCQM HPC JupyterHub Cloud Computing Portal Slurm + Quantum ESPRESSO

LIVE PRODUCTION DEPLOYMENT



Centralized Authentik SSO & Identity OAuth2 / OIDC + OpenLDAP

INTERDISCIPLINARY VALUE PROPOSITION FOR ACADEMY PROGRAM

Physics Rigor

First-principles analytical problem formulation, statistical mechanics modeling, and deep mathematical grounding for complex systems.

HPC Systems

Hardware-level clustering, low-latency InfiniBand/RDMA fabric, Slurm partition orchestration, and bare-metal enterprise Linux administration.

Web Software

End-to-end fullstack architecture (Laravel 11, Filament, Docker), REST/OIDC security, DNS deliverability, and automated business workflows.

CANDIDATE COMMITMENT & ACADEMY READINESS

Albert Andersen brings a rare combination of scientific research discipline and hands-on systems implementation. Capable of translating abstract algorithmic and mathematical models into deployed, high-throughput web and computing environments with production-grade reliability.

Direct Verification & Full Interactive Portfolio: alberttandersen.github.io • Tel / WhatsApp: +62 813-1704-5100